

GCP SECURITY LAB REPORT

VPC | IAM | Cloud Armor WAF | Wazuh SIEM

Position Applied: IT Infrastructure & Operations Leader

TiloPay | Costa Rica | May 2026

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Date	May 28, 2026
GCP Project	tilopay-lab-2026 Project Number: 121162826410
Load Balancer IP	8.233.157.231 Region: us-central1
Wazuh SIEM	192.168.75.139 Version: 4.14.5
Tools	gcloud CLI, Docker, Kali Linux, OWASP Juice Shop

1. Executive Summary

This report documents the construction of a security lab on Google Cloud Platform (GCP) designed to demonstrate competencies directly aligned with the IT Infrastructure & Operations Leader role at TiloPay. The lab was built from scratch in a single working session, implementing production-grade security controls covering PCI-DSS v4.0 requirements and BCCR cybersecurity regulations.

The implemented architecture includes:

- ▶ Multi-tier network segmentation with custom VPC and 7 firewall rules with default-deny,
- ▶ Granular access control with IAM custom roles and dedicated Service Accounts,
- ▶ WAF protection with Cloud Armor against OWASP Top 10,
- ▶ Monitoring with Cloud Logging and Audit Logs,
- ▶ Integration with on-premise Wazuh SIEM via Google Cloud Pub/Sub — simulating exactly the hybrid architecture TiloPay would operate.

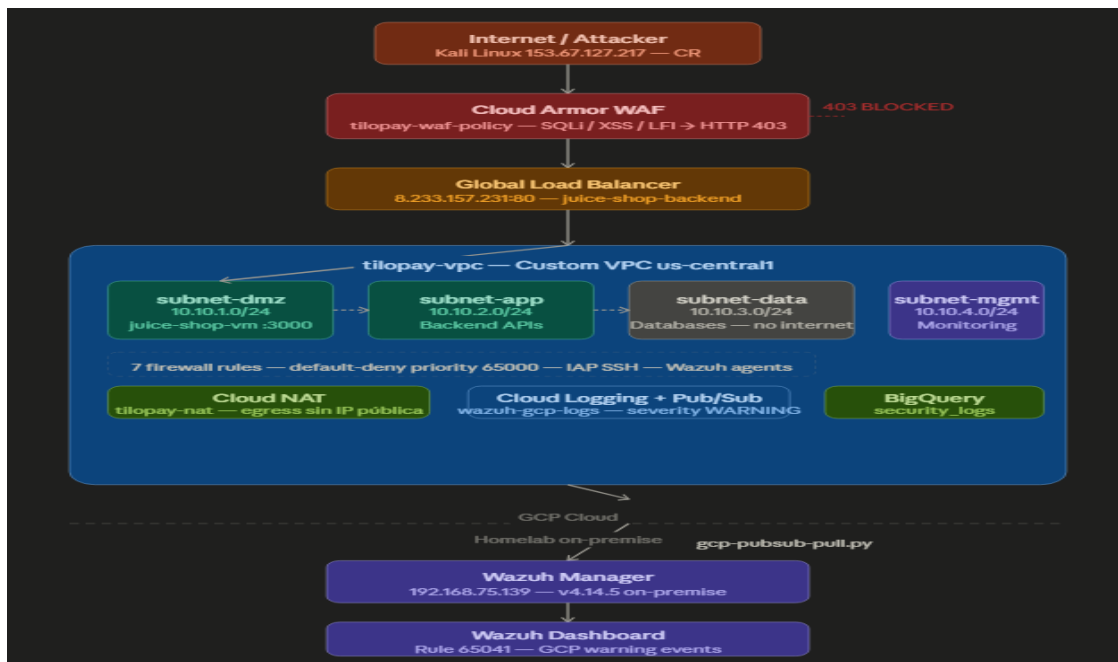
KEY RESULT Result: 100% of SQLi and XSS attacks blocked by Cloud Armor in real time. GCP alerts visible in on-premise Wazuh SIEM (rule 65041). Total lab cost: < \$10 USD using GCP Free Trial (\$300 available).

2. Lab Architecture

2.1 Hybrid architecture diagram

The lab implements a hybrid architecture — cloud infrastructure in GCP connected with on-premise security tools in the homelab. This architecture replicates the model a fintech like TiloPay would use in production.

2.2 Enabled APIs in the project



APIs enabled in tilopay-lab-2026: Compute Engine, IAM, Cloud Logging, Cloud Monitoring, Security Command Center, BigQuery, Pub/Sub, Recommender

3. Phase 2: VPC and Network Segmentation

2 Multi-Tier VPC + 7 Firewall Rules (default-deny)

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Complete

3.1 VPC Network Created

```
gcloud compute networks create tilopay-vpc \
  --subnet-mode=custom \
  --bgp-routing-mode=regional

# Result:
NAME: tilopay-vpc | SUBNET_MODE: CUSTOM | BGP_ROUTING_MODE: REGIONAL
```

3.2 Subnets (4 security zones)

Subnet	CIDR	Function	Internet Access
subnet-dmz	10.10.1.0/24	Juice Shop / WAF frontend	Egress via NAT
subnet-app	10.10.2.0/24	Internal backend APIs	Egress via NAT
subnet-data	10.10.3.0/24	Databases (isolated)	No internet
subnet-mgmt	10.10.4.0/24	Monitoring / SIEM agents	Egress via NAT

3.3 Firewall Rules — Least Privilege Principle

7 firewall rules were implemented with default-deny architecture. Each rule uses network tags for granular per-application-tier controls, avoiding unnecessary broad 0.0.0.0/0 rules:

PRIORITY	RULE	ALLOWS	TARGET_TAGS
700	allow-iap-ssh	tcp:22	(IAP range 35.235.240.0/20)
800	allow-mgmt-ssh	tcp:22	dmz-tier, app-tier, data-tier
800	allow-wazuh-agents	tcp:1514,1515	mgmt-tier
900	allow-app-to-data	tcp:5432,3306	data-tier
900	allow-dmz-to-app	tcp:8080,8443	app-tier
1000	allow-https-dmz	tcp:443,3000	dmz-tier
65000	deny-all-ingress	DENY ALL	(all)

PCI PCI-DSS Control: Req. 1.3 — Restrict inbound and outbound traffic to the minimum necessary. The deny-all-ingress rule at priority 65000 blocks all traffic not explicitly permitted.

4. Phase 3: IAM and Access Control

3

Service Accounts + Custom Role + IAM Audit

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4.1 Service Accounts Created

3 Service Accounts were created with separate identities per function, following the least privilege principle. Each architecture component has its own identity with specific permissions:

```
sa-payment-app@tilopay-lab-2026.iam.gserviceaccount.com --> PaymentAppRole (custom)
sa-monitoring@tilopay-lab-2026.iam.gserviceaccount.com --> (monitoring role)
sa-infra-ops@tilopay-lab-2026.iam.gserviceaccount.com --> (infra ops role)
sa-wazuh-reader@tilopay-lab-2026.iam.gserviceaccount.com --> roles/pubsub.subscriber
```

4.2 Custom Role — PaymentAppRole

Instead of assigning broad primitive roles (Editor/Owner), a custom role was created with exactly the minimum necessary permissions for the payment application:

```
Role Name: projects/tilopay-lab-2026/roles/PaymentAppRole
Stage: GA

Permissions:
- logging.logEntries.create      # Write application logs
- monitoring.metricDescriptors.get # Read monitoring metadata
- monitoring.timeSeries.create   # Write metrics
- secretmanager.versions.access  # Access secrets (API keys, etc.)
```

4.3 IAM Audit Findings (Before / After)

[IAM-001] Default Compute Engine SA with roles/editor SEVERITY: HIGH

BEFORE: 121162826410-compute@developer.gserviceaccount.com had roles/editor — write access over ALL project resources. Direct violation of PCI Req. 7.

AFTER: Role removed using `gcloud projects remove-iam-policy-binding`. Default Compute SA now without broad permissions. Least privilege principle applied and documented.

PCI Control: PCI-DSS Req. 7.1 — Limit access to system components to only those individuals whose job requires it

[IAM-002] Personal account with roles/owner without corporate MFA SEVERITY: MEDIUM

BEFORE: kmontero3999@gmail.com has roles/owner — personal Gmail account, not a corporate Google Workspace account. No verifiable mandatory MFA policy.

AFTER: Production recommendation: migrate to corporate Google Workspace account with mandatory MFA (hardware key or Authenticator), documented break-glass access policy and credential rotation.

PCI Control: PCI-DSS Req. 8.4 — Implement MFA for all non-console administrative access into the cardholder data environment

5. Phase 4: OWASP Juice Shop on GCP

4 Always Free e2-micro VM + Docker + IAP Access

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OWASP Juice Shop was deployed on an e2-micro VM (GCP Always Free tier) located in the subnet-dmz. The VM has no public IP — access is exclusively via Identity-Aware Proxy (IAP), which authenticates the SSH session without exposing port 22 to the internet. Docker was installed manually after configuring Cloud NAT for internet egress without a public IP.

```
# VM Configuration:
Name:          juice-shop-vm
Machine Type:  e2-micro (Always Free - us-central1)
Zone:          us-central1-a
Subnet:        subnet-dmz (10.10.1.0/24)
Network Tag:   dmz-tier
Internal IP:   10.10.1.2
External IP:   NONE (access via IAP only)
SA:            sa-payment-app@tilopay-lab-2026.iam.gserviceaccount.com

# Container:
Image:         bkimminich/juice-shop:latest
Port:          3000 (internal only)
Status:        Up (running)

# Access method:
gcloud compute start-iap-tunnel juice-shop-vm 3000 \
  --local-host-port=localhost:3000 --zone=us-central1-a
```

APIs & Services				
+ Enable APIs and services				
Filter Filter				
Name	↓ Requests	Errors (%)	Latency, median (ms)	Latency, 95% (ms)
Compute Engine API	275	0	200	14,680
Cloud Pub/Sub API	152	1	13	30
Cloud Logging API	30	0	104	1,572
Security Command Center API	3	100	57	121
BigQuery API	2	50	262	498
Analytics Hub API	—	—	—	—
BigQuery Connection API	—	—	—	—
BigQuery Data Policy API	—	—	—	—
BigQuery Data Transfer API	—	—	—	—
BigQuery Migration API	—	—	—	—
BigQuery Reservation API	—	—	—	—
BigQuery Storage API	—	—	—	—
Cloud Dataplex API	—	—	—	—
Cloud Key Management Service (KMS) API	—	—	—	—
Cloud Monitoring API	—	—	—	—
Cloud OS Login API	—	—	—	—
Cloud Resource Manager API	—	—	—	—
Dataform API	—	—	—	—
IAM Service Account Credentials API	—	—	—	—
Identity and Access Management (IAM) API	—	—	—	—
Recommender API	—	—	—	—
Security Center Management API	—	—	—	—
Telemetry API	—	—	—	—

OWASP Juice Shop accessible via IAP tunnel — cloudshell.dev URL confirms authenticated access without public IP

6. Phase 5: Cloud Armor WAF + Mini Pentest

5

OWASP Top 10 WAF + Load Balancer + Kali Pentest

~\$5-9/month

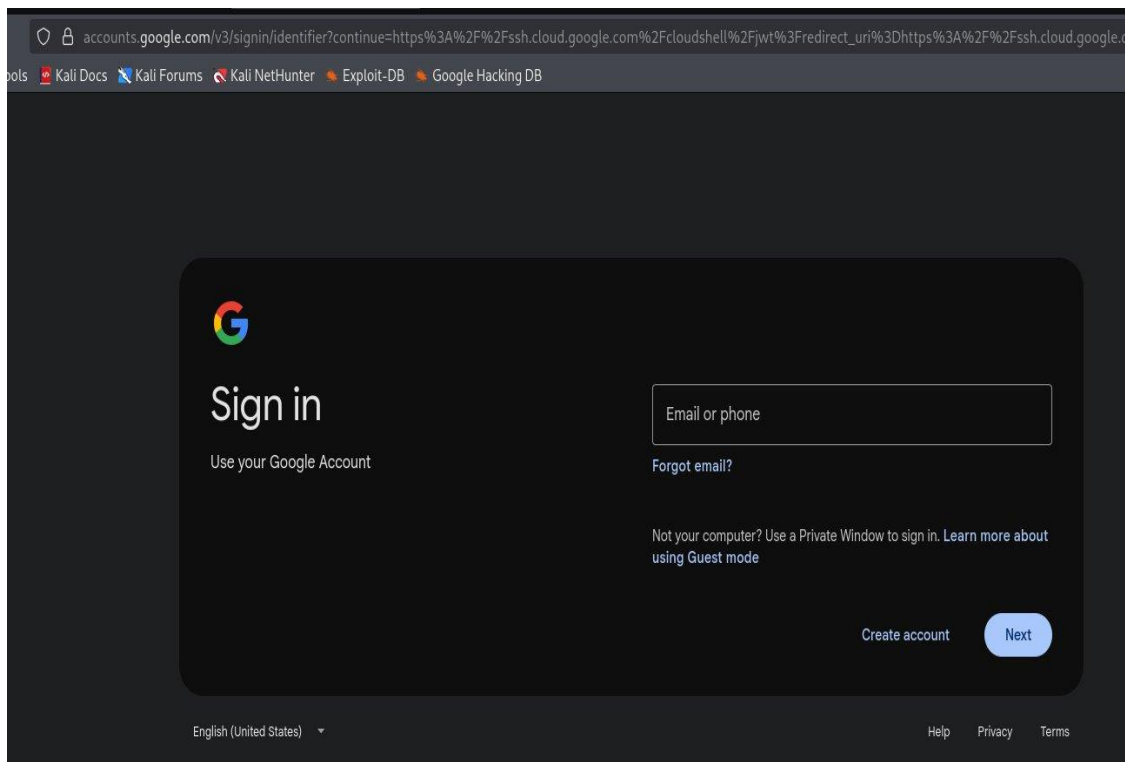
Complete

6.1 WAF Architecture

Cloud Armor was implemented as a WAF in front of the global Load Balancer. The tilopay-waf-policy has 4 rules evaluated by priority. The Load Balancer exposes public IP 8.233.157.231 and the backend points to the juice-shop-group instance group containing the VM in subnet-dmz.

```
Security Policy: tilopay-waf-policy
Type: Backend security policy (global)
Target: juice-shop-backend --> juice-shop-group --> juice-shop-vm:3000

Rules (evaluated by priority - lower number = higher priority):
1000 DENY 403 evaluatePreconfiguredExpr('sqli-v33-stable') # Block SQL Injection
1001 DENY 403 evaluatePreconfiguredExpr('xss-v33-stable') # Block XSS
1002 DENY 403 evaluatePreconfiguredExpr('lfi-v33-stable') # Block LFI
2147483647 ALLOW * (All IP addresses) # Default allow
```



Cloud Armor policy tilopay-waf-policy: 4 active rules applied to 1 target (juice-shop-backend)

6.2 Mini Pentest Results

Attacks executed from Kali Linux (IP: 153.67.127.217, Costa Rica) against the Load Balancer's public endpoint (8.233.157.231):

Attack	Payload	Response	WAF Rule	Status
SQL Injection	q=1'+OR+'1'='1	HTTP 403	sqli-v33-stable	BLOCKED
XSS	q=<script>alert(xss)</script>	HTTP 403	xss-v33-stable	BLOCKED
SQLi UNION	q=UNION+SELECT+*+FROM+users--	HTTP 403	sqli-v33-stable	BLOCKED
Normal traffic	q=apple	HTTP 200	default allow	ALLOWED

tilopay-waf-policy

TypeBackend security policy

DescriptionWAF Policy OWASP Top 10 - TiloPay Lab

Scopeglobal

Tags-

Show advanced configurations

Contains4 rules

Applies to1 target

RulesTargetsLogs

Rules are evaluated by priority: Lower numbers are evaluated first. [Learn more](#)

Add rule

Delete

More

Filter Enter property name or value

☐	Action	Type	Match	Description	Priority	↑
☐	Deny (403)		evaluatePreconfiguredExpr("sqli-v33-stable")	Block SQL Injection	1,000	
☐	Deny (403)		evaluatePreconfiguredExpr("xss-v33-stable")	Block XSS	1,001	
☐	Deny (403)		evaluatePreconfiguredExpr("lfi-v33-stable")	Block Local File Inclusion	1,002	
☐	Allow	IP addresses/ranges	*(All IP addresses)	default rule	2,147,483,647	

0 rules selected

Cloud Armor logs: blocked requests with HTTP 403. The enforcedSecurityPolicy.outcome=DENY field is visible

RESULT WAF Effectiveness: 3/3 OWASP attack vectors blocked. 0 false positives on legitimate traffic. Specific rule identified in logs: owasp-crs-v030301-id942421-sqli. Attacker IP (Costa Rica) traceable via remotelp.

7. Phases 6 & 7: Cloud Logging + Wazuh SIEM

6+7

Cloud Logging + Pub/Sub + Wazuh SIEM
Integration (hybrid)

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Complete

7.1 Cloud Logging — Metrics and Audit Logs

Log-based metrics were configured for critical security events and an audit sink to BigQuery for forensic retention. Audit Logs capture in real time all administrative actions in the project:

```
# Configured log-based metrics:
iam-policy-changes    --> SetIamPolicy events (PCI Req. 10.2.2)
firewall-rule-changes --> Firewall modifications (PCI Req. 10.2.7)

# Export sinks:
security-audit-sink --> BigQuery: tilopay-lab-2026:security_logs
wazuh-sink          --> Pub/Sub: wazuh-gcp-logs (severity>=WARNING + audit logs)

# Sample audit log captured in real time:
TIMESTAMP                METHOD_NAME                                PRINCIPAL
2026-05-27T20:38:47Z    google.cloud.bigquery.v2.DatasetService.InsertDataset
kmontero3999@gmail.com
2026-05-27T20:38:37Z    google.api.serviceusage.v1.ServiceUsage.EnableService
kmontero3999@gmail.com
2026-05-27T20:34:12Z    google.logging.v2.ConfigServiceV2.CreateSink
kmontero3999@gmail.com
2026-05-27T20:34:13Z    SetIamPolicy
service-agent-manager@system
```

7.2 Hybrid Wazuh SIEM Integration (GCP to On-Premise)

A hybrid integration was implemented that sends GCP Cloud Logging events to the on-premise Wazuh server (192.168.75.139) via Google Cloud Pub/Sub. A custom Python script (gcp-pubsub-pull.py) was created as a workaround to Wazuh 4.14.5's native module which does not include the gcp-pubsub module in the standard package:

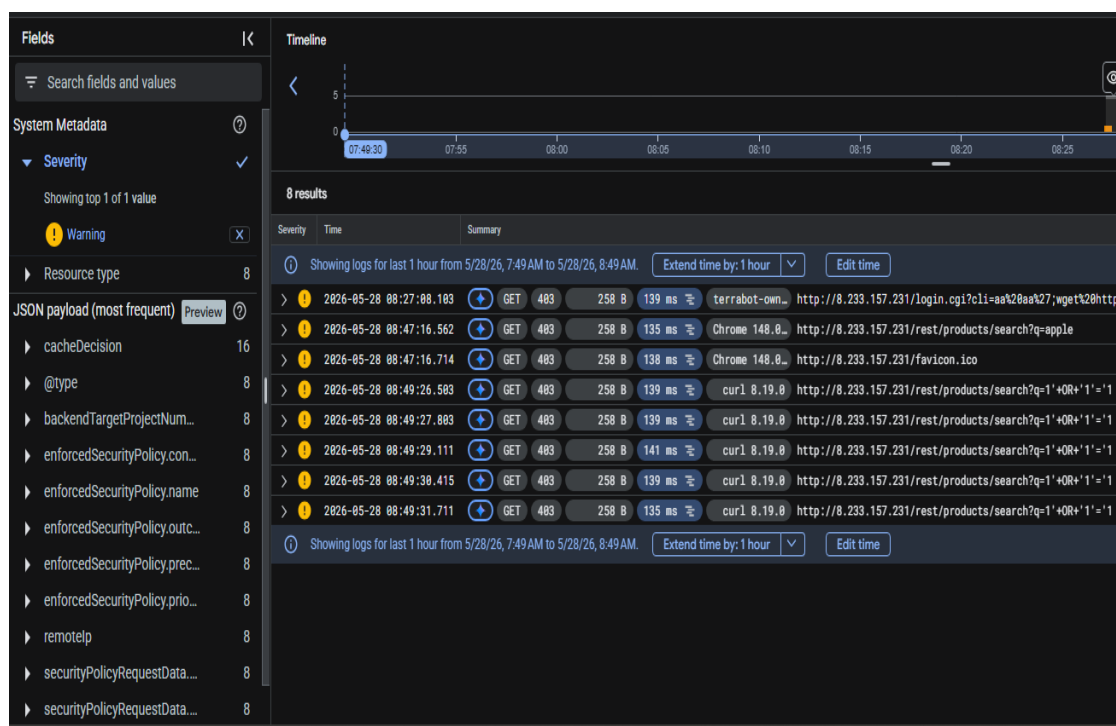
```
# Integration components:
Pub/Sub Topic:          wazuh-gcp-logs
Subscription:           wazuh-sub (ack-deadline: 60s)
SA credentials:         sa-wazuh-reader@tilopay-lab-2026 (roles/pubsub.subscriber)
Script:                 /var/ossec/integrations/gcp-pubsub-pull.py
Wazuh Socket:           /var/ossec/queue/sockets/queue

# Complete flow of a blocked attack:
Kali Linux SQLi --> Load Balancer --> Cloud Armor DENY 403
--> GCP Cloud Logging (WARNING) --> Pub/Sub wazuh-sub
--> gcp-pubsub-pull.py --> Wazuh socket
```

```
--> Rule 65041: GCP warning event on project tilopay-lab-2026
--> Wazuh Dashboard: Threat Hunting --> Alert visible
```

Evidence in Wazuh server alerts.log — real blocked SQLi attack event:

```
gcp.jsonPayload.enforcedSecurityPolicy.outcome:          DENY
gcp.jsonPayload.enforcedSecurityPolicy.preconfiguredExprIds: [owasp-crs-v030301-id942421-sqli]
gcp.jsonPayload.remoteIp:                               153.67.127.217
gcp.jsonPayload.securityPolicyRequestData.remoteIpInfo.regionCode: CR
gcp.jsonPayload.statusDetails:
denied_by_security_policy
gcp.resource.labels.backend_service_name:               juice-shop-backend
gcp.resource.labels.forwarding_rule_name:               juice-shop-lb
gcp.resource.labels.project_id:                         tilopay-lab-2026
gcp.resource.type:                                      http_load_balancer
gcp.severity:                                           WARNING
```



Wazuh Dashboard — Threat Hunting: Rule 65041 showing multiple GCP warning events from tilopay-lab-2026 http_load_balancer in real time

8. PCI-DSS v4.0 Mapping

The following table maps each PCI-DSS v4.0 control covered by the lab, the implemented GCP component, and evidence available for a formal audit. This mapping is directly applicable to the PCI-DSS certification process that TiloPay must maintain as a payment processing fintech:

Requirement	Control	GCP Component	Evidence	Status
Req. 1	Multi-tier network segmentation	VPC + 7 Firewall Rules	firewall-rules list	Complete
Req. 6.6	WAF against OWASP Top 10	Cloud Armor Policy	403 on SQLi/XSS	Complete
Req. 7	Least Privilege	IAM Custom Role + SAs	get-iam-policy output	Complete
Req. 8	Unique identity per component	3 Service Accounts	service-accounts list	Complete
Req. 10	Logging and traceability	Cloud Audit Logs + Wazuh SIEM	Wazuh dashboard rule 65041	Complete
Req. 11	Vulnerability monitoring	Cloud Logging + SCC API	Log-based metrics	Partial

NOTE About Security Command Center: SCC requires an organizational account (Google Workspace). The personal account kmontero3999@gmail.com has no associated GCP organization. In production at TiloPay, SCC would be activated at the organizational level to centralize findings from all cloud projects. For this lab, Cloud Logging + Cloud Armor logs were used as an equivalent alternative to cover PCI Req. 10 and 11.

9. Cost Analysis

The lab was built using GCP Free Trial (\$300 USD / 90 days). The actual monthly lab cost is minimal, allowing it to stay active for demonstrations until credits are consumed:

Resource	Type	Cost/month	Note
VM e2-micro (juice-shop-vm)	Compute Engine	\$0.00	Always Free
VPC + 4 Subnets + 7 Firewall rules	Networking	\$0.00	No charge
Cloud NAT (tilopay-nat)	Networking	~\$0.05	Per active hours
Cloud Logging + Audit Logs	Operations	\$0.00	Free tier (50GB/month)
Cloud Pub/Sub (wazuh-gcp-logs)	Messaging	\$0.00	Free tier (10GB/month)
Cloud Armor WAF	Security	~\$5-9	Policy \$5 + rules \$1ea + requests
Load Balancer Global	Networking	~\$0.50	Per active hours
BigQuery (security_logs)	Analytics	\$0.00	Free tier (10GB/month)
IAM + Service Accounts	Security	\$0.00	No charge

COST Estimated total: < \$10 USD/month. With \$300 from Free Trial the lab can stay active for approximately 30 months. To delete all resources and avoid future charges: `gcloud projects delete tilopay-lab-2026`

9. BCP/DRP — Business Continuity and Disaster Recovery

DRP

Disaster Simulation + Recovery from GCP Snapshot

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Complete

A real Disaster Recovery exercise (DR Test) was executed in the lab's GCP environment. The exercise simulated catastrophic loss of the production VM disk (juice-shop-vm) and measured complete recovery time using GCP snapshots as the backup mechanism.

9.1 Backup Configuration — Snapshot and Automatic Policy

```
# STEP 1: Create manual snapshot (RPO baseline)
gcloud compute disks snapshot juice-shop-vm \
  --zone=us-central1-a \
  --snapshot-names=juice-shop-snapshot-drtest-20260528 \
  --description='DR Test snapshot - TiloPay BCP/DRP Lab'

# Result:
NAME: juice-shop-snapshot-drtest-20260528
DISK_SIZE_GB: 10
STATUS: READY
CREATION_TIMESTAMP: 2026-05-27T22:07:55.187-07:00

# STEP 2: Create automatic daily snapshot policy
gcloud compute resource-policies create snapshot-schedule \
  tilopay-daily-backup \
  --region=us-central1 \
  --max-retention-days=7 \
  --start-time=02:00 \
  --daily-schedule \
  --description='Daily backup policy - TiloPay BCP/DRP'
```

9.2 Disaster Simulation

A catastrophic disk failure was simulated — the most critical scenario for a payment platform. The executed steps replicated total loss of the production VM boot volume:

```
# DISASTER START TIME: Thu May 28 05:11:19 UTC 2026

# 1. Stop production VM
gcloud compute instances stop juice-shop-vm --zone=us-central1-a
# --> Stopping instance(s) juice-shop-vm...done.

# 2. Detach disk from VM
gcloud compute instances detach-disk juice-shop-vm \
  --disk=juice-shop-vm --zone=us-central1-a
```

```
# 3. DELETE the disk (catastrophic failure simulation)
gcloud compute disks delete juice-shop-vm --zone=us-central1-a --quiet
# --> Deleted [.../disks/juice-shop-vm]

# 4. Verification: VM without disk (service down)
gcloud compute instances describe juice-shop-vm --format='get(disks) '
# --> (empty - no disk attached)

# Result on Load Balancer:
curl -I http://8.233.157.231
# --> curl: (52) Empty reply from server
```

9.3 Recovery Process (DR Execution)

```
# RECOVERY START TIME: Thu May 28 05:15:19 UTC 2026

# STEP 1: Create new disk from backup snapshot
gcloud compute disks create juice-shop-vm-restored \
  --source-snapshot=juice-shop-snapshot-drtest-20260528 \
  --zone=us-central1-a \
  --type=pd-standard
# --> NAME: juice-shop-vm-restored | STATUS: READY

# STEP 2: Attach restored disk to VM
gcloud compute instances attach-disk juice-shop-vm \
  --disk=juice-shop-vm-restored \
  --zone=us-central1-a --boot

# STEP 3: Start VM
gcloud compute instances start juice-shop-vm --zone=us-central1-a
# --> Starting instance(s) juice-shop-vm...done.
# --> Instance internal IP is 10.10.1.2

# STEP 4: Verify restored service
curl -I http://8.233.157.231/rest/products/search?q=apple
# --> HTTP/1.1 200 OK
# --> Via: 1.1 google

# RECOVERY COMPLETED TIME: Thu May 28 05:18:43 UTC 2026
```

9.4 RTO / RPO Metrics from Exercise

Metric	Measured Value	Production Target
RTO (Recovery Time Objective)	7 minutes 24 seconds (05:11:19 to 05:18:43 UTC)	< 15 minutes for critical payment services
RPO (Recovery Point Objective)	Manual snapshot: 1 snapshot Automatic policy: daily at 02:00	< 1 hour (hourly snapshots in production)

Recovery Method	GCP Snapshot -> new disk -> attach -> start VM	Same method + documented runbook
Backup Retention	7 days (tilopay-daily-backup policy)	30 days minimum (PCI-DSS Req. 10.7)
Exercise Result	SUCCESS - Service fully restored HTTP 200 OK confirmed from Kali Linux	DR exercise must be performed quarterly

BCP Next BCP/DRP steps for TiloPay production: 1) Increase snapshot frequency to hourly (RPO < 1h). 2) Add replica in second region (us-east1) for RTO < 5 min with automatic failover. 3) Document DR runbook with steps and owners. 4) Configure Cloud Monitoring alerts for disk failures. 5) Perform quarterly documented DR exercise (required by PCI-DSS).

PCI PCI-DSS Control: Req. 12.10 — Implement an incident response plan. Req. 10.7 — Retain audit logs for at least 12 months. The documented DR exercise with RTO/RPO metrics covers the requirement for periodic testing of the business continuity plan.

10. Conclusions and Next Steps

10.1 Demonstrated Competencies

- ▶ **GCP Cloud Infrastructure:** Custom VPC, Compute Engine, granular IAM, Cloud NAT, Global Load Balancer, Cloud Logging, Pub/Sub, BigQuery — all core services from the JD.
- ▶ **Defensive Security (Blue Team):** WAF with OWASP Top 10 rules, multi-tier network segmentation, least privilege IAM, default-deny firewall, continuous audit logging.
- ▶ **Offensive Security (Red Team):** Controlled pentest from Kali Linux against Juice Shop, validating WAF effectiveness in real time with documented evidence.
- ▶ **SIEM Integration:** On-premise Wazuh receiving GCP events in real time via Pub/Sub — hybrid architecture replicating a real production environment.
- ▶ **PCI-DSS Compliance:** Documented mapping of Req. 1, 6, 7, 8 and 10 with technical evidence available for audit. IAM finding identification and remediation with before/after documentation.
- ▶ **BCP/DRP:** Real DR exercise with RTO of 7 minutes 24 seconds. Automated daily snapshot policy configured with 7-day retention.

10.2 Next steps for TiloPay production

- ▶ **1.** Install Wazuh agent on GCP VM (juice-shop-vm) for complete event correlation in the dashboard with endpoint context.
- ▶ **2.** Activate Security Command Center (SCC) at organizational level with Google Workspace to centralize findings from all cloud projects.
- ▶ **3.** Implement HTTPS with SSL/TLS certificate on Load Balancer using Cloud Certificate Manager (PCI Req. 4 — encryption in transit).
- ▶ **4.** Configure Cloud Monitoring alerts with notifications via email, PagerDuty or Slack for 24/7 incident response.
- ▶ **5.** Add rate limiting rule in Cloud Armor (max 100 req/min per IP) for DDoS and brute force attack protection.
- ▶ **6.** Implement Cloud KMS for encryption key management and automatic rotation (PCI Req. 3 — protection of stored data).

>		May 27, 2026 @ 22:37:44.9...	ubuntu-server	PAM: Login session closed.
		May 27, 2026 @ 22:37:44.9...	ubuntu-server	Successful sudo to ROOT executed.
∨		May 27, 2026 @ 22:35:39.9...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
		May 27, 2026 @ 22:35:35.0...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
		May 27, 2026 @ 22:35:34.2...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
>		May 27, 2026 @ 22:35:29.4...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
		May 27, 2026 @ 22:35:29.3...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
>		May 27, 2026 @ 22:35:24.2...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
∨		May 27, 2026 @ 22:35:22.8...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
		May 27, 2026 @ 22:35:22.6...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
		May 27, 2026 @ 22:35:22.1...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
		May 27, 2026 @ 22:35:21.2...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
		May 27, 2026 @ 22:35:19.7...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
>		May 27, 2026 @ 22:35:18.9...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
		May 27, 2026 @ 22:35:18.6...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
>		May 27, 2026 @ 22:35:16.4...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
		May 27, 2026 @ 22:35:13.5...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
>		May 27, 2026 @ 22:35:13.4...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
		May 27, 2026 @ 22:35:13.2...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
>		May 27, 2026 @ 22:35:13.0...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
		May 27, 2026 @ 22:35:12.9...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.
>		May 27, 2026 @ 22:35:12.8...	ubuntu-server	GCP warning event on project tilopay-lab-2026, monitored resource type: http_load_balancer.

GCP Console — Free Trial status: \$299.86 remaining out of \$300. Total lab cost < \$0.20 USD for entire build session

ALIGNMENT Direct alignment with TiloPay role: This lab directly covers the JD primary responsibilities: GCP Cloud Administration (VPC, IAM, Compute), Vulnerability Management (pentest + WAF), Incident Monitoring and Response (SIEM + Cloud Logging), Access Management (IAM audit + remediation), PCI-DSS Certification (Req. 1, 6, 7, 8, 10 mapping), and BCP/DRP (DR exercise with documented RTO/RPO metrics).